All Gujarat Vankar Samaj

સંપૂર્ણ ડિજિટલ પરિચય અને મેટ્રિમોનિયલ પ્લેટફોર્મ (Official Platform Documentation)

Live URL: https://allgujaratvankarsamaj.com **Status:** Production Ready **Version:** 1.0.10+11

Updated: October 2026

1 Executive Summary & Platform Mission

All Gujarat Vankar Samaj is a premier, community-driven digital matrimonial, socio-cultural, and career networking platform engineered specifically for the **Vankar Samaj** community across Gujarat, India, and the international diaspora.

The portal bridges rich cultural traditions with cutting-edge technology:

- **100% Genuine Matrimonial Matchmaking:** Candidate profiles with mandatory government ID proof verification, verified contact flags, and dual-side image capture.
- **AI-Powered Identity Verification:** Live multimodal vision checks using Google Gemini AI, cross-verifying candidate Name and Date of Birth against uploaded ID documents to prevent identity fraud.
- **Cultural Lineage & Directory:** Interactive repository of all Vankar Parganas, Talukas, and Native Villages (મૂળ ગામ અને મોસાળ ગામ).
- **Community Pride & Honors:** Recognition hubs honoring Samaj Super Stars, Samaj Ratna awardees, Pavan Prernadata (mentors), and verified Government Employees (IAS, Police, Teachers, Bank officers).
- **Social Upliftment & Services:** Samaj Sevaks emergency network, education scholarships, competitive exam coaching, and verified business advertisements.

2 High-Level System Architecture & Tech Stack

Frontend (Multiplatform Web & Mobile)

Flutter 3.47 / Dart 3.13
Single responsive codebase delivering Flutter Web (WASM/CanvasKit/HTML) and native Android/iOS builds. State managed via **Riverpod 2.6** with **GoRouter 14.8** deep linking.

Backend API (Enterprise NestJS)

NestJS 10.4 / Node.js 20+
High-concurrency modular backend implementing RESTful APIs, Swagger documentation, JWT authentication, and automated error handling.

Database & ORM

Prisma ORM 5.22 + MySQL / PostgreSQL
Fully normalized relational database schema with

Artificial Intelligence & Vision Engine

Google Gemini 1.5 Flash Vision AI
Sub-second multimodal document analysis validating

type-safe queries, automatic migrations, and indexed foreign key relationships.

government IDs, extracting bilingual Gujarati/English text, and validating candidate name/DOB.

Component	Technology	Key Role
Web Gateway	Apache + PHP 8.3 Reverse Proxy	High-speed Unix Domain Socket (`backend.sock`) proxy with automated watchdog auto-restart (`api_proxy.php`).
Authentication	Passport-JWT & Bcrypt	Dual identifier login (Phone or Email), salted password hashing, 7-day token expiration, and RBAC permissions.
Storage Engine	Dual-Tier Storage Service	Public media in `/uploads` (avatars, galleries) and private secured storage in `/private_uploads` (Aadhaar, Govt IDs) protected by RBAC streaming.
OTP Service	Fast2SMS & SMTP Gateway	4-digit / 6-digit OTP delivery with automated 40-second cooldown timer and resend lockout guards.

3 Complete Feature Guide

3.1 Authentication & Registration

- **Dual-Identifier Login (`/login`):** Login via 10-digit mobile number OR email address. Session auto-refreshed via `/auth/me` to prevent profile field loss.
- **OTP Verification with 40-Second Cooldown (`/register`):**
  - Real-time countdown timer: `Resend in 40s` .
  - Bilingual guidance alert displayed on expiration: *"OTP કોડ નથી મળ્યો? ફરી કોડ મોકલવા માટે ઉપર 'ફરી કોડ મોકલો' બટન દબાવો."*
  - Phone & Email verification synchronization with user account.
- **Password Recovery & Update:** In-app password change modal and self-service credential updates.

3.2 Matrimonial Profiles (Create & Edit)

- **Personal Profile:** First Name, Last Name, Gender, Date of Birth (calendar picker), Marital Status (Never Married, Divorced, Widowed), Blood Group, Mother Tongue.
- **Community Lineage:** Sub-caste, Pargana selection, Native Village (મૂળ ગામ), Mother's Village (મામાનું ગામ), Taluka, District, State, Country.
- **Education & Career:** Qualifications (Doctor, Engineer, CA/CS, MBA, Graduate, etc.), Employment Sector (Govt Central, Govt State, Corporate, Business, Abroad), Designation, Annual Income.
- **Family Details:** Father's Name & Occupation, Mother's Name & Occupation, Siblings, Guardian contact.
- **Photo Gallery:** High-resolution primary profile picture + 2 to 5 lifestyle gallery photos with client-side 5 MB enforcement.
- **3-Tier Privacy Settings:** Independent visibility settings for Mobile Number, Email, and Address (`બધા માટે` / `પરિવારજનો` / `માત્ર મને`).

3.3 AI-Powered Government ID Document Verification

> **Strict AI Verification Pipeline:** When a user attaches their Aadhaar Card, PAN Card, Voter ID, or Driving License, the system triggers live Google Gemini Vision AI to read the document.

- **Name Cross-Check:** Extracts name in English and Gujarati; intelligently accounts for Indian honorifics (*bhai*, *kumar*, *ben*, *devi*) and father's middle name.
- **Date of Birth Cross-Check:** Validates full `DD/MM/YYYY` dates or Aadhaar birth years (`Year of Birth: YYYY`).
- **Strict Zero-False-Positive Visual State:**
  - The card border and status banner **never** turn green based solely on file selection.

- Green badge `✓ AI Verified` and status `✓ ઓળખપત્ર AI દ્વારા ચકાસાયેલ છે` strictly require real AI verification approval.
- Mismatched or fake documents (such as uploading someone else's ID) are **rejected immediately and blocked from attaching**.

4 Community, Lineage & Directory Portals

Portal Route	Module Name	Description & Purpose
`/pargana-overview`	**Pargana & Village Directory**	Interactive directory listing all Parganas across Gujarat, registered talukas, villages, and community leader contact info.
`/government-employees`	**Government Employees Cell**	Roster of community civil servants, police officers, railway officials, bank managers, and educators with verified badges.
`/samaj-super-stars`	**Samaj Super Stars**	Hall of fame celebrating high achievers, entrepreneurs, national awardees, and sports personalities.
`/samaj-ratna`	**Samaj Ratna Awards**	Lifetime honor roll recognizing distinguished elders and community benefactors.
`/pavan-prernadata`	**Pavan Prernadata**	Historical leaders, spiritual guides, and inspiring mentors of the Vankar Samaj.
`/samaj-services`	**Samaj Services & Sevaks**	Directory of volunteer social workers, emergency blood donors, medical aid coordinators, and funeral assistance.
`/education`	**Education & Career Hub**	GPSC/UPSC competitive exam coaching materials, student scholarship opportunities, and hostel directories.
`/family-details`	**Family & Lineage Archive**	Digital tree capturing family lineage, native roots, and genealogical ties.
`/success-stories`	**Success Stories**	Couples united through the platform with wedding pictures and family testimonials.
`/statistics`	**Community Analytics**	Demographic charts covering education levels, district distributions, and employment ratios.

5 Discovery, Matchmaking & Communication

- **Quick & Advanced Search (`/search` , `/advanced-search`):** Multi-criteria search by age range, height, sub-caste, pargana, village, education degree, sector, and income.
- **Candidate Profile Details (`/candidate-profile-details`):** Rich biodata view with picture zoom, horoscope details, family info, and one-tap *Express Interest* (રસ દાખવો).
- **Match Recommendations (`/match`):** Daily mutual compatibility suggestions based on age, education, and pargana preferences.
- **Real-Time 1-to-1 Chat (`/chat/:conversationId`):** Private messaging between mutual matches with image and document exchange.

6 Administration & Moderation Panel

- **Executive KPI Dashboard:** Real-time counts for Total Profiles, Pending ID Verifications, Active Members, and Daily Traffic.
- **Verification Queue (`/admin/verifications`):** Side-by-side inspection of Front and Back ID photos, AI confidence score, and one-click Approve / Reject with custom message.
- **OTP Telemetry:** Live metrics tracking SMS delivery success rate, carrier latencies, and failure trends.
- **User Moderation:** Suspend fraudulent users, review community reports, and feature verified profiles.

> **Document Certification:** This technical documentation accurately reflects the live production architecture, Flutter 3.47 frontend codebase, and NestJS backend implementation of **All Gujarat Vankar Samaj** as of October 2026.